

Aerodynamic Analysis and Drag Optimization of the Orbit Biplane Tail-Sitter

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Introduction

In the past few years, the aerospace industry has seen an increase in interest in Vertical Take-Off and Landing (VTOL) aircrafts and this has led to the development of new configurations such as the Biplane Tail-Sitters which has both the hovering capability of rotary-wing aircrafts as well as the cruise efficiency of fixed-wing aircrafts. A biplane tailsitter is a VTOL configured aircraft with a biplane wing arrangement to get the required aerodynamic performance while keeping the VTOL capabilities during take-off and landing. Unlike tilt-rotors or tilt-wings, a biplane tailsitter does not require a single extra motor, servo, or hinge to transition flight modes. The entire aircraft simply pitches forward by changing the differential thrust of its four fixed rotors.

The tailsitter aircrafts come up with unique aerodynamic challenges due to their unconventional geometry and operating conditions. Unlike the normal fixed-wing aircrafts, tailsitters have to operate efficiently in vertical take-off and horizontal cruise while doing a transition between these two. The biplane geometry introduces problems in the wing-wing interactions and the structural connectors joining the two wings. This could significantly affect the aerodynamic parameters of the aircraft.

Background

The Orbit is an existing VTOL configured biplane tailsitter aircraft, developed in Helicopter Lab, at IIT Kanpur. It has been completely designed and manufactured in the lab and it is capable of vertical take-off and landing with cruise efficiency.

The main purpose of this aircraft is aerial surveillance, terrain mapping, small scale cargo or package delivery. Such aircrafts are advantageous as they do not require a runway for take-off or landing and are efficient in cruise. But the efficiency of the baseline configuration can be affected by factors such as parasitic and induced drag, wing-wing interactions. Hence, a detailed aerodynamic analysis is required to search for opportunities of improvement.

The main objective of the project is to analyze the aerodynamic characteristics of the baseline configuration of Orbit biplane tailsitter and look for new design modifications that can improve lift while reducing drag and makes the overall model more effective. The work focuses on understanding the lift and drag behaviour of the aircraft and identify the parts and sections that are mainly affecting the overall drag and implement the modifications.

Methodology

Learning and Preliminary Aerodynamic Studies

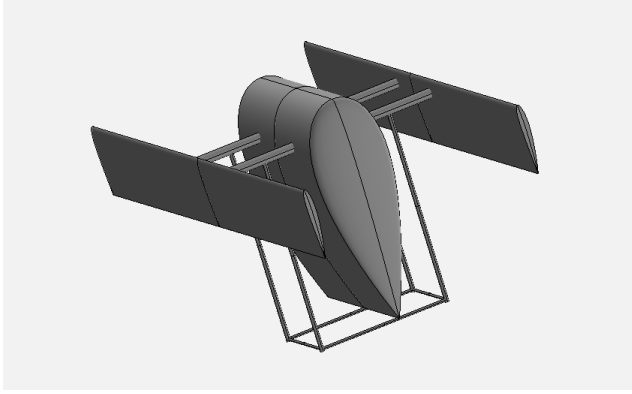
The project begins with learning to use low fidelity softwares such as OpenVSP and FlightStream. Many different wing and aircraft geometries are created to get used to working with these tools and understand geometries, mesh generation and solver configuration procedures. The aerodynamic analyses are done for various geometries to study the effect of the geometric parameters on aerodynamic performance.

The simulations are run on both OpenVSP and FlightStream, and aerodynamic coefficients such as the lift coefficient (C_L), induced drag coefficient (C_{D_i}), and parasitic drag coefficient (C_{D_o}) are calculated. The results from OpenVSP and FlightStream are then compared with each other for reliability and consistency of the calculations.

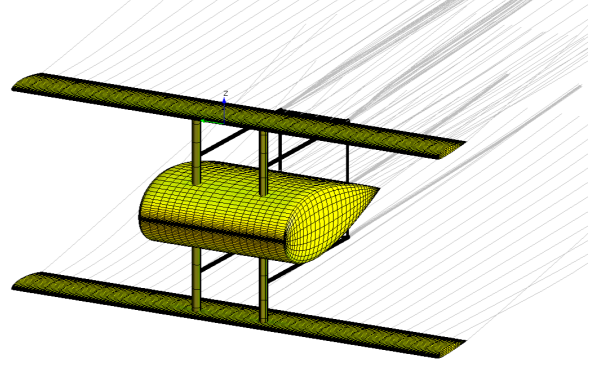
Orbit Biplane Geometry Development and Analysis

A detailed geometry of the Orbit biplane tasilsitter is developed with exact dimensions in OpenVSP. More attention is given to the wings and the fuselage. The complete geometry serves as the baseline configuration.

The Orbit geometry is first analyzed using VSPAERO in OpenVSP and the geometry is imported to FlightStream for improved analysis. Key aerodynamic parameters are evaluated and compared across both software applications. The results from both softwares are compared to see consistency and are further compared with available literature to validate the simulations.



(a) Orbit biplane tail-sitter geometry created in OpenVSP.



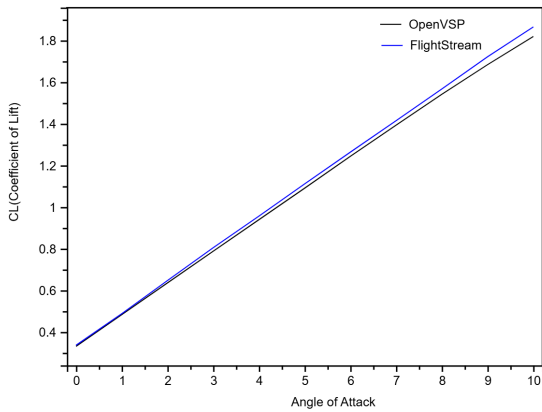
(b) Aerodynamic analysis setup and surface mesh in FlightStream.

Figure 1: The aircraft geometry was developed in OpenVSP and subsequently analyzed in VSPAERO and FlightStream for aerodynamic performance evaluation.

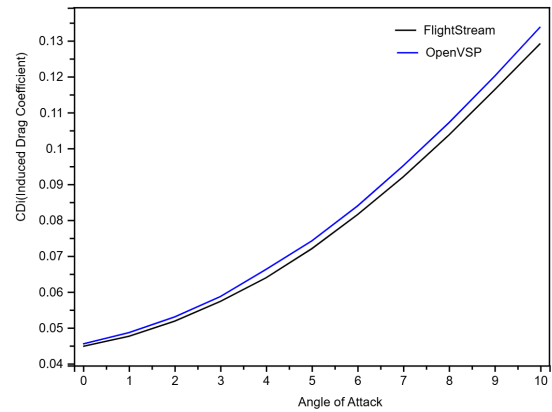
Validation and Preliminary Results

The main objective is to gain confidence on the working softwares before moving onto high fidelity tools. we got the validation through the comparison between the results obtained from OpenVSP and FlightStream as well as through available data. The main focus is to compare the lift coefficient(C_L) and induced drag coefficient(C_{D_i}).

The results obtained from OpenVSP and FlightStream showed similar aerodynamic trends over a range of angle of attack(0° - 10°). At lower angles of attack(0° - 4°), the results from both are almost overlapping. But as the angle of attack increased, there is a gradual separation in trends between the two softwares. The separation between the results of both the softwares goes upto 5%.



(a) C_L variation with angle of attack.



(b) C_{D_i} variation with angle of attack.

Figure 2: Comparison of aerodynamic coefficients predicted by OpenVSP and FlightStream for the Orbit biplane tail-sitter.

Future Work

The next phase of the project will be focused on more detail analysis of Orbit biplane tailsitter using OpenVSP and FlightStream. The next part is to develop a slightly more accurate model, focusing more on the structural connectors joining the wing.

Different design changes in overall geometry such as wing geometry, airfoil changes, taper ratio, structural connectors design and configurations, will be implemented. The simulations will be performed using OpenVSP and FlightStream on new design configurations and the aerodynamic parameters will be evaluated. The results for each simulation will be compared to the baseline configuration and check whether the change has improved the efficiency. Similarly, all the results will be compared and will look for the best possible design with high efficiency.

The overall goal is to perform complete aerodynamic analysis and optimize overall drag of the aircraft while keeping in mind of its operational requirements. This work will also provide foundation for any future work and study of VTOL and biplane tailsitter aircrafts.

Keywords: Tail-sitter aircraft, Aerodynamic analysis, OpenVSP, FlightStream, Drag optimization, Lift Coefficient(C_L), Induced Drag Coefficient(C_{D_i}), VTOL.

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